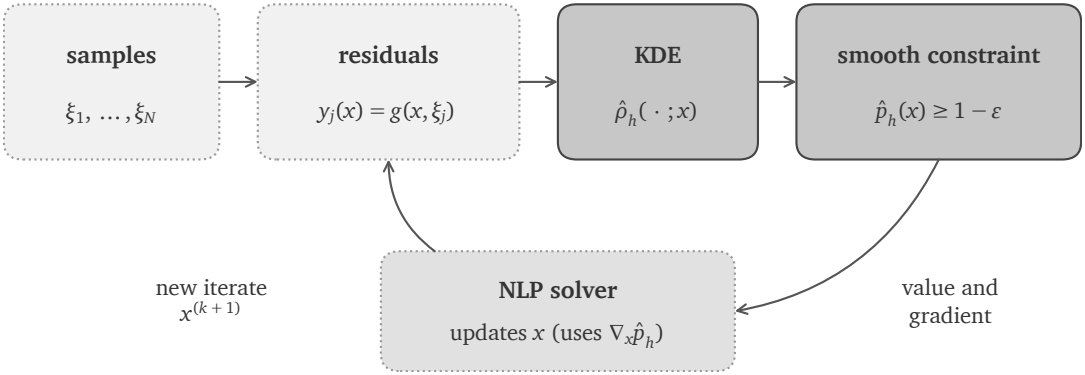
samples
$\xi_1, \ldots, \xi_N$
residuals
$y_j(x) = g(x, \xi_j)$
KDE
$\hat{\rho}_h(\,\cdot\,; x)$
smooth constraint
$\hat{p}_h(x) \geq 1 - \varepsilon$
new iterate
$x^{(k+1)}$
NLP solver
updates x (uses $\nabla_x \hat{p}_h$)
value and
gradient